

Effectiveness of Dapagliflozin and Sitagliptin Fixed Dose Combination in Indian Patients with Type 2 Diabetes Mellitus in Real World Settings: Age Stratified-Analysis from a Retrospective Study

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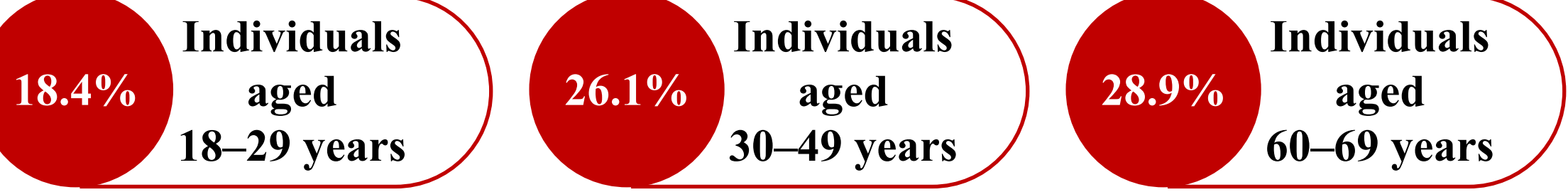
INTRODUCTION & OBJECTIVE

Burden

- The incidence of T2DM is rapidly increasing resulting in significant diabetes-related complications ¹

Global prevalence ^{1,2}	Indian prevalence ^{3,4}
537 million	101 million
Increased by 90.5% from 1990 and 2021	Increased from 5.5% in 1990 to 9.6% in 2021

- According to the National Noncommunicable Disease Monitoring Survey conducted in India, impaired fasting glucose is observed in ⁵



- This necessitates the need for early diagnosis and treatment for preventing complications, improving quality of life, and ensuring longevity ⁵
- Early and tight glycemic control can help to mitigate this risk of diabetes-related complications and prevent beta-cell decline ⁶
- In addition, > 75% of Indian patients have uncontrolled diabetes (HbA1c ≥ 7%), requiring sequential addition of medications with multiple tablets ^{7,8}

Importance of FDC

- Combination therapy provides effective control of blood glucose and do not increase the risk of hypoglycemia or other AEs ⁹

FDC ^{8,10}	Multiple OADs ⁸
- Reduces pill burden, dosing frequency, and cost - Improved adherence - Complementary mechanism of action - Sustained glycemic control	- Increases pill burden and dosing frequency - Poor adherence leading to inadequate glycemic control

Synergistic effect of Dapagliflozin and Sitagliptin FDC

- Dapagliflozin and Sitagliptin FDC can benefit patients with T2DM due to its ability to address multiple pathophysiologies of T2DM, potentially simplifying therapy. ^{11,12,13} Additionally in Indian patients,
- DPP4i is widely used, well-tolerated, and has increased efficacy due to an increase in DPP4 enzyme activity ^{12,14,15}
- SGLT2i, besides glycemic control, offers unique benefits such as weight reduction, improvement in β-cell function, beneficial effect on cardiovascular and renal system ¹³

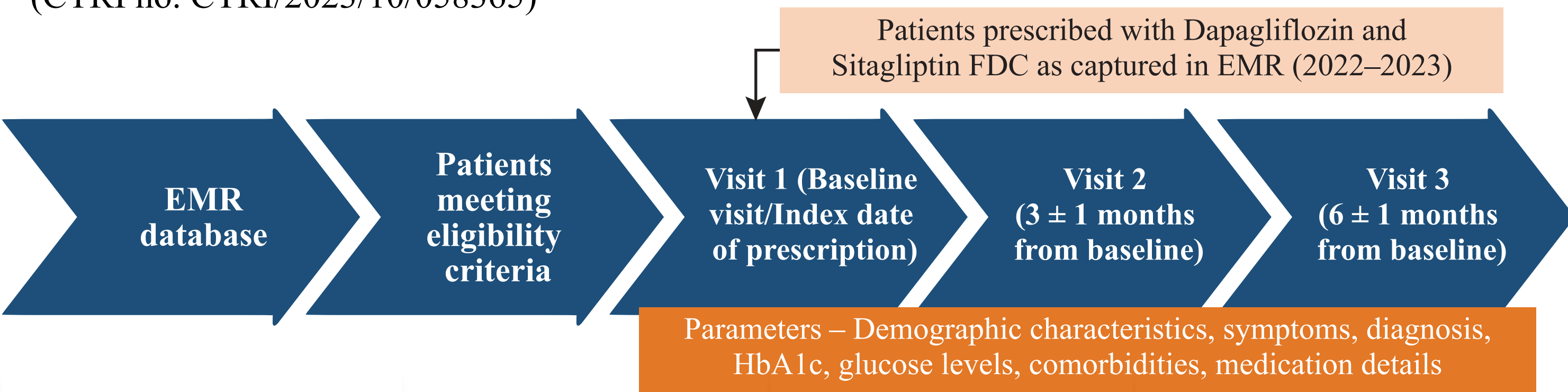
Synergistic effect of Dapagliflozin and Sitagliptin FDC	
SGLT2i (Dapagliflozin)	- Improved β-cell function ¹³ - Increased urinary glucose excretion ¹⁶ - Increased glucagon and EGP/HGP ¹⁶ - Weight reduction ¹³ - Beneficial effect on cardiovascular and renal system ¹³
DPP4i (Sitagliptin)	- Increased efficacy due to an increase in DPP4 enzyme activity in Asian Indian population ¹² - Slows degradation of incretin hormones ¹⁶ - Inhibit glucagon and reduces EGP/HGP ¹⁶ - Blocks glucose reabsorption of distal renal tubule to exert natriuresis effects ¹⁶ - Well-tolerated ¹⁴

Need and objective of real-world evidence study

- RWE is essential to understand the generalizability and applicability of Dapagliflozin and Sitagliptin FDC, in clinical practice and to help clinicians in making informed decisions about patient treatment ¹⁷
- Limited real-world data exist on the effectiveness of Dapagliflozin and Sitagliptin FDC in India ¹⁷
- A retrospective, observational, EMR-based study was conducted to assess the real-world effectiveness and usage pattern of Dapagliflozin and Sitagliptin FDC in Indian patients. The results of age stratified analysis is presented here.

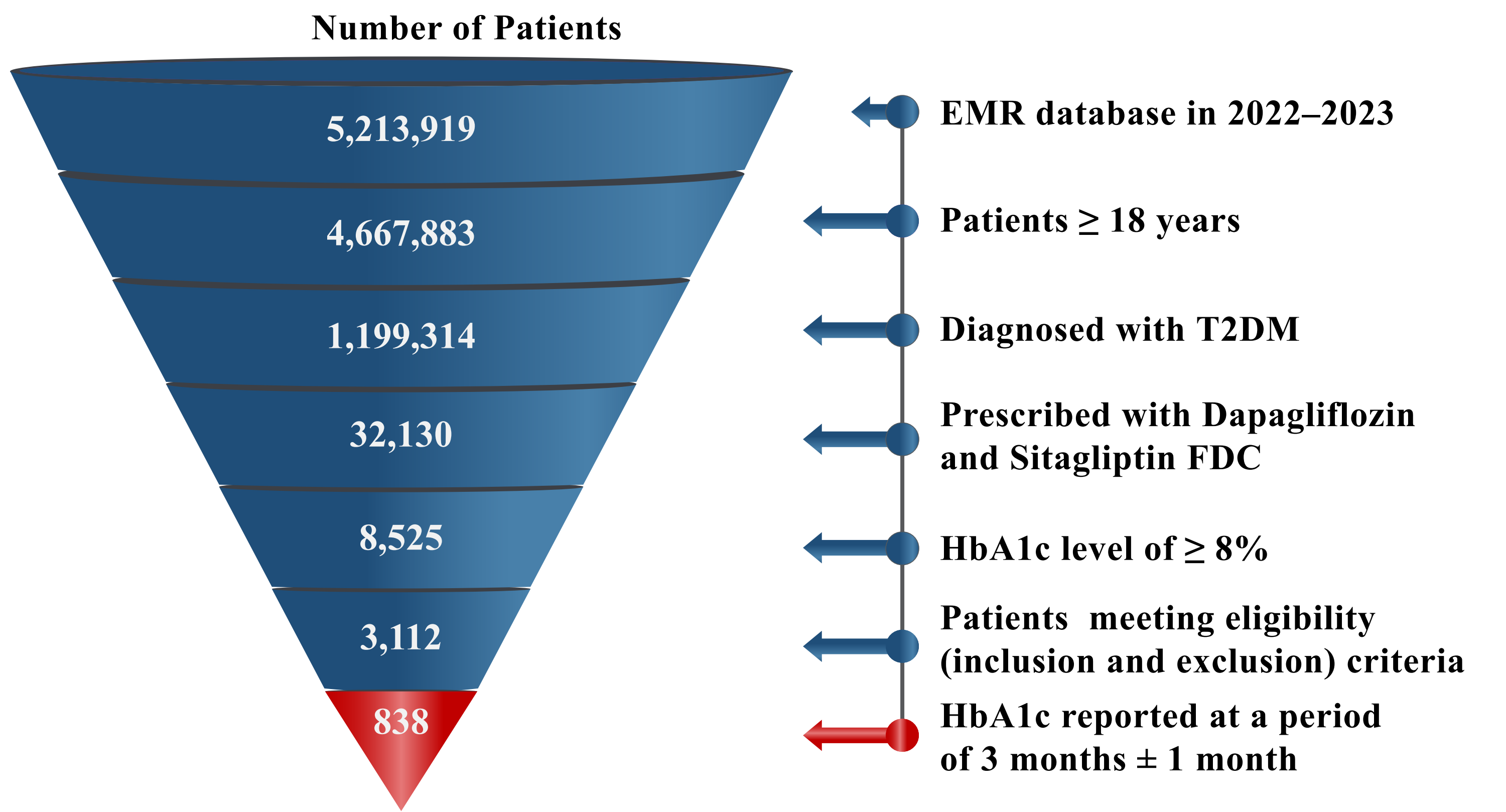
METHODS

- In this study, aggregated and anonymized EMR data from the patients meeting the eligibility criteria was retrieved and analyzed. Study was approved by Independent Ethics Committee and registered on CTRI (CTRI no. CTRI/2023/10/058365)



Inclusion criteria	Exclusion criteria	Primary endpoint	Key secondary endpoints
- Patients with - Age ≥ 18 years - Diagnosed with T2DM - HbA1c ≥ 7% - Prescribed Dapagliflozin and Sitagliptin FDC in any visit other than baseline visit on the EMR platform - EMR data captured for more than one visit	- T1DM - Patients on insulin or any other injectable antidiabetic medication - Patients with incomplete medical records with respect to the required study parameters	Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥ 8% at baseline	- Mean change in FBG and PPBG level from baseline to 3 months in patients with HbA1c ≥ 8% at baseline - Mean change in body weight from baseline to 3 months in patients with HbA1c > 8% with BMI > 23 kg/m² and > 25 kg/m² at baseline

RESULTS



Used for age-stratified analysis

Group 1 Age ≤ 50 years (N = 299)	Group 2 Age 51–60 years (N = 281)	Group 3 Age 61–70 years (N = 198)	Group 4 Age > 70 years (N = 60)
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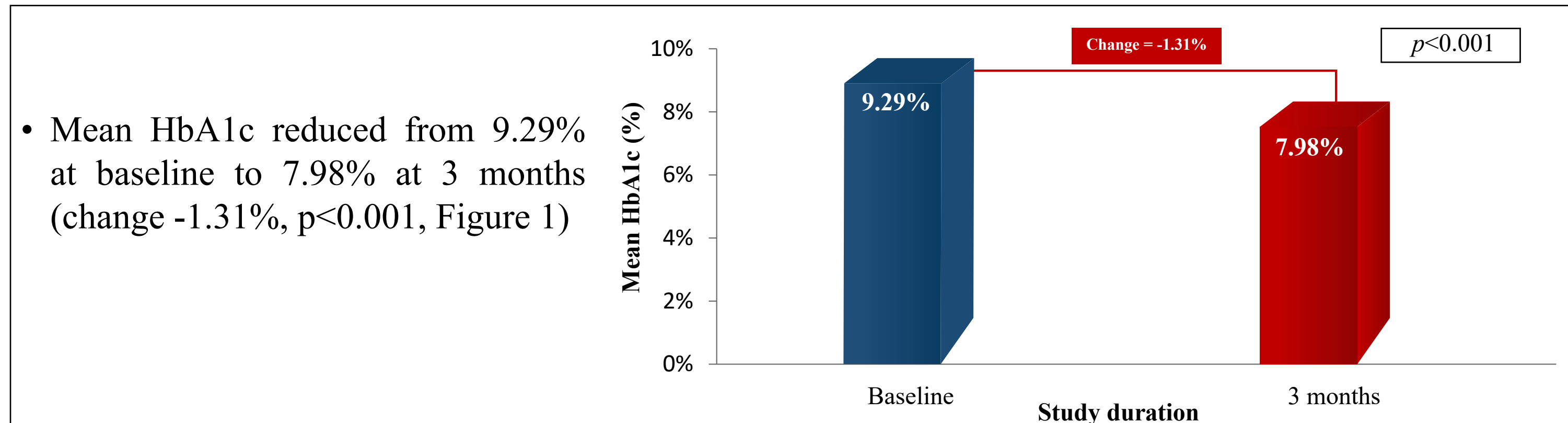


Figure 1: Mean change in HbA1c among different age groups

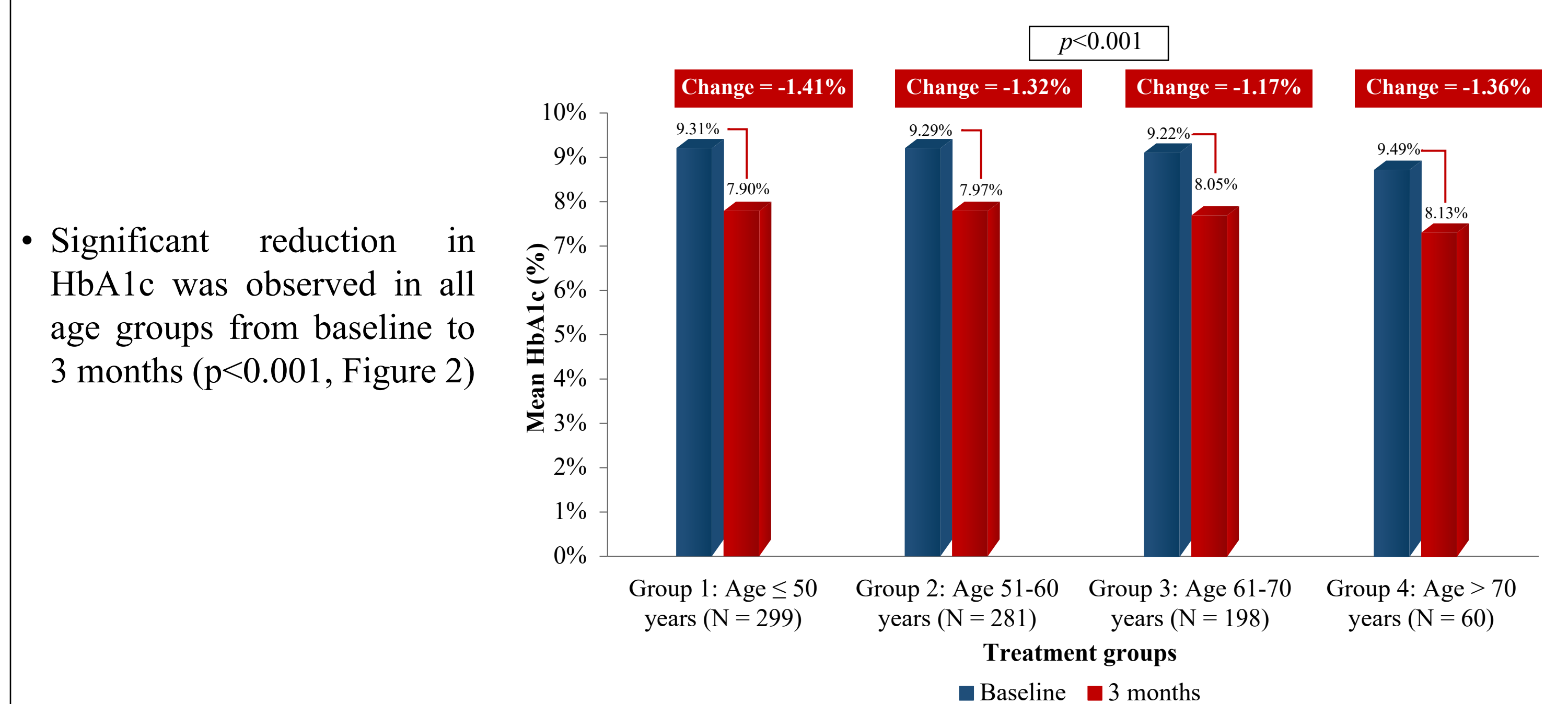


Figure 2: Mean change in overall HbA1c from baseline to 3 months

- Significant reduction in FBG and PPBG was observed in all age groups from baseline to 3 months (p<0.001, Tables 1 and 2)

Table 1: Mean change in FBG (mg/dL) among different age groups

	Group 1: Age ≤ 50 years (N = 251)	Group 2: Age 51–60 years (N = 236)	Group 3: Age 61–70 years (N = 170)	Group 4: Age > 70 years (N = 53)
Baseline	175.67	166.83	161.77	155.61
3 months	135.27	136.19	128.26	123.66
Change (mg/dL)	-38.4	-30.64	-33.51	-31.95

Table 2: Mean change in PPBG (mg/dL) among different age groups

	Group 1: Age ≤ 50 years (N = 251)	Group 2: Age 51–60 years (N = 205)	Group 3: Age 61–70 years (N = 154)	Group 4: Age > 70 years (N = 58)
Baseline	175.67	254.08	240.52	246.10
3 months	200.02	201.32	197.63	202.52
Change (mg/dL)	-52.17	-52.76	-42.89	-43.58

CONCLUSION

- Dapagliflozin and Sitagliptin FDC was effective in overall glycemic control in patients with T2DM
- Dapagliflozin and Sitagliptin FDC effectively reduced HbA1c, FBG, and PPBG in patients with T2DM across all age groups within 3 months, indicating effective glycemic control and reduced blood sugar levels
- This highlights utility of Dapagliflozin and Sitagliptin FDC in managing T2DM irrespective of age and can be considered as part of effective management in patients with T2DM in real-world setting

Disclosure of interest: Ritesh Agrawala: None declared, Supratik Bhattacharyya: None declared, Prakadeesh Bharathi B: Author is full time employee of Sun Pharma Laboratories Ltd., Vinay Kudrigikar: Author is full time employee of Sun Pharma Laboratories Ltd., Prachi Ahire: Author is full time employee of Sun Pharma Laboratories Ltd., Shruti Dharmadhikari: Author is full time employee of Sun Pharma Laboratories Ltd., Chintan Khandhedia: Author is full time employee of Sun Pharma Laboratories Ltd., Neeraj Markandeywar: Author is full time employee of Sun Pharma Laboratories Ltd., Amey Mane: Author is full time employee of Sun Pharma Laboratories Ltd., Suyog Mehta: Author is full time employee of Sun Pharma Laboratories Ltd.

Acknowledgment: Authors would like to thank HealthPlix for providing data source.

Abbreviations: AE, adverse event; EGP, endogenous glucose production; DPP4i, dipeptidyl peptidase 4 inhibitor; EMR, electronic medical record; FDC, fixed dose combination; FBG, fasting blood glucose; HbA1c, glycated hemoglobin; HGP, hepatic glucose production; OAD, oral antidiabetic drug; PPBG, postprandial blood glucose; SGLT2i, sodium-glucose cotransporter-2 inhibitor; T1DM, type 1 diabetes; T2DM, type 2 diabetes
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